# SPARQL 1.2 Query Results XML Format




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# Abstract

# Status of This Document

*This section describes the status of this document at the time of its publication. A list of current W3C publications and the latest revision of this technical report can be found in the W3C technical reports index at https://www.w3.org/TR/.*

This specification is published by the RDF Star Working Group as part of the update of specifications for format and errata.

This document was published by the RDF-star Working Group as a Working Draft using the Recommendation track.

Publication as a Working Draft does not imply endorsement by W3C and its Members.

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This document is governed by the 03 November 2023 W3C Process Document.

## § Set of Documents

This document is one of twelve SPARQL 1.2 documents produced by the RDF-star Working Group.

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# § 1. Introduction

The *SPARQL 1.2 Query Language* defines several *Query Result Forms* (SPARQL Query section 10). This document defines a SPARQL Results Document that encodes the variable binding query results from `SELECT` queries (SPARQL Query section 10.2) and boolean query results from `ASK` queries (SPARQL Query section 10.5) in XML.

There are two other results formats which follow a similar design but do not use XML: *SPARQL 1.2 Query Results JSON Format* and *SPARQL 1.2 Query Results CSV and TSV Formats*.

# § 2. Definition

> **Definition:** SPARQL Results Document
>
> A **SPARQL Results Document** is an XML document that is valid with respect to either the RELAX NG XML Schema or the W3C XML Schema in Section 4.

## § 2.1 Document Element

The SPARQL Results Document begins with `sparql` document element in the `http://www.w3.org/2005/sparql-results#` namespace, written as follows:

```
<?xml version="1.0"?>
<sparql xmlns="http://www.w3.org/2005/sparql-results#"
  xmlns:its="http://www.w3.org/2005/11/its"
  its:version="2.0">
 ...
</sparql>
```

Inside the `sparql` element are two sub-elements, `head` and a results element (either `results` or `boolean`) which must appear in that order.

If no literals with base direction appear in the results, the `sparql` document element may be simplified as follows.

```
<?xml version="1.0"?>
<sparql xmlns="http://www.w3.org/2005/sparql-results#">
 ...
</sparql>
```

> NOTE
>
> Different values of `its:version` are allowed.

## § 2.2 Header

The `head` element is the first child element of the `sparql` element.

For a variable binding query result, `head` must contain a sequence of elements describing the set of Query Variable names in the Solution Sequence (here called query results).

The order of the variable names in the sequence is the order of the variable names given to the argument of the `SELECT` statement in the SPARQL query. If `SELECT *` is used, the order of the names is undefined.

Inside the `head` element, the ordered sequence of variable names chosen are used to create empty child elements `variable` with the variable name as the value of an attribute `name` giving a document like this:

```
<?xml version="1.0"?>
<sparql xmlns="http://www.w3.org/2005/sparql-results#"
  xmlns:its="http://www.w3.org/2005/11/its"
  its:version="2.0">

  <head>
    <variable name="x"/>
    <variable name="hpage"/>
    <variable name="name"/>
    <variable name="mbox"/>
    <variable name="blurb"/>
  </head>
...
</sparql>
```

For a boolean query result, no elements are required inside `head` and `variable` must not be present.

For any query result, `head` may also contain `link` child elements with an `href` attribute containing a relative URI that provides a link to some additional metadata about the query results. The relative URI is resolved against the in-scope base URI which is usually the query results format document URI. `link` elements must appear after any `variable` elements that are present.

```
<?xml version="1.0"?>
<sparql xmlns="http://www.w3.org/2005/sparql-results#"
  xmlns:its="http://www.w3.org/2005/11/its"
  its:version="2.0">

  <head>
    ...
    <link href="metadata.rdf"/>
  </head>
...
</sparql>
```

## § 2.3 Results

The second child-element of `sparql` must appear after `head` and is either `results` or `boolean`. It is written even if the query results are empty.

### § 2.3.1 Variable Binding Results

The `results` element contains the complete sequence of query results.

For each Query Solution in the query results, a `result` child-element of `results` is added giving a document like:

```
<?xml version="1.0"?>
<sparql xmlns="http://www.w3.org/2005/sparql-results#"
  xmlns:its="http://www.w3.org/2005/11/its"
  its:version="2.0">
  ...  head ...

  <results>
    <result>...
    </result>
    <result>...
    </result>
    ...
  </results>

</sparql>
```

Each `result` element corresponds to one *Query Solution* in a result and contains child elements (in no particular order) for each *Query Variable* that appears in the solution. It is used to record how the query variables bind to RDF Terms.

Each binding inside a solution is written as an element `binding` as a child of `result` with the query variable name as the value of the `name` attribute. So for a result binding two variables *x* and *hpage* it would look like:

```
<?xml version="1.0"?>
<sparql xmlns="http://www.w3.org/2005/sparql-results#"
  xmlns:its="http://www.w3.org/2005/11/its"
  its:version="2.0">
  <head>
    <variable name="x"/>
    <variable name="hpage"/>
  </head>

  <results>
    <result>
      <binding name="x"> ... </binding>
      <binding name="hpage"> ... </binding>
    </result>

    <result>
      <binding name="x"> ... </binding>
      <binding name="hpage"> ... </binding>
    </result>
    ...
  </results>

</sparql>
```

The value of a query variable binding, which is an *RDF Term*, is included as the content of the `binding` as follows:

**RDF URI Reference *U***

```
<binding><uri>U</uri></binding>
```

**RDF Literal *S***

```
<binding><literal>S</literal></binding>
```

**RDF Literal *S* with language *L* without base direction**

```
<binding><literal xml:lang="L">S</literal></binding>
```

**RDF Literal *S* with language *L* with base direction *L***

```
<binding><literal xml:lang="L" its:dir="B">S</literal></binding>
```

**RDF Typed Literal *S* with datatype URI *D***

```
<binding><literal datatype="D">S</literal></binding>
```

**Blank Node label *I***

```
<binding><bnode>I</bnode></binding>
```

**Triple Term, with subject *S*, predicate *P*, object *O***

```
<binding>
  <triple>
    <subject>S</subject>
    <predicate>P</predicate>
    <object>O</object>
  </triple>
</binding>
```

If, for a particular solution, a variable is *unbound*, no `binding` element for that variable is included in the `result` element.

*S*, *P*, and *O* in Triple Terms are encoded recursively, using the same format, without the enclosing `<binding>` tag

**Note:** The blank node label *I* is scoped to the result set XML document and need not have any association to the blank node label for that RDF Term in the query graph.

An example of a query solution encoded in this format is as follows:

```
<?xml version="1.0"?>
<sparql xmlns="http://www.w3.org/2005/sparql-results#"
  xmlns:its="http://www.w3.org/2005/11/its"
  its:version="2.0">

  <head>
    <variable name="x"/>
    <variable name="hpage"/>
    <variable name="name"/>
    <variable name="age"/>
    <variable name="mbox"/>
    <variable name="friend"/>
  </head>

  <results>

    <result>
      <binding name="x">
        <bnode>r2</bnode>
      </binding>
      <binding name="hpage">
        <uri>http://work.example.org/bob/</uri>
      </binding>
      <binding name="name">
        <literal xml:lang="en">Bob</literal>
      </binding>
      <binding name="age">
        <literal datatype="http://www.w3.org/2001/XMLSchema#integ
      </binding>
      <binding name="mbox">
        <uri>mailto:bob@work.example.org</uri>
      </binding>

    ...
  </results>

</sparql>
```

An example of a query solution that includes triple terms is as follows:

```
<?xml version="1.0"?>
<sparql xmlns="http://www.w3.org/2005/sparql-results#"
  xmlns:its="http://www.w3.org/2005/11/its"
  its:version="2.0">

  <head>
    <variable name="x"/>
    <variable name="name"/>
    <variable name="triple"/>
  </head>

  <results>

    <result>
      <binding name="x">
        <bnode>r2</bnode>
      </binding>
      <binding name="name">
        <literal xml:lang="en">Bob</literal>
      </binding>
      <binding name="triple">
        <triple>
            <subject>
                <uri>http://example.org/alice</uri>
            </subject>
            <predicate>
                <uri>http://example.org/name</uri>
            </predicate>
            <object>
                <literal datatype="http://www.w3.org/2001/XMLSche
            </object>
        </triple>
      </binding>
    </result>

    ...
  </results>

</sparql>
```

As an alternative to including the `xml:its` declaration in every result set, the namespace can be declared on specific elements as needed:

```
<?xml version="1.0"?>
<sparql xmlns="http://www.w3.org/2005/sparql-results#">
  <head>
    <variable name="animal"/>
  </head>
  <results>
    <result>
      <binding name="animal">
        <literal xmlns:its="http://www.w3.org/2005/11/its" its:ve
                 xml:lang="ar" its:dir="rtl">قطة</literal>
      </binding>
    </result>
    <result>
      <binding name="animal">
        <literal xml:lang="en">cat</literal>
      </binding>
    </result>
    <result>
      <binding name="animal">
        <literal xmlns:its="http://www.w3.org/2005/11/its" its:ve
                 xml:lang="fr" its:dir="ltr">chat</literal>
      </binding>
    </result>
  </results>
</sparql>
```

### § 2.3.2 Boolean Results

A boolean result is written as the element content of a `boolean` child-element of the `sparql` element directly after a `head`, containing either `true` or `false` as follows:

```
<?xml version="1.0"?>
<sparql xmlns="http://www.w3.org/2005/sparql-results#">
  ...  head ...

  <boolean>true</boolean>

</sparql>
```

## § 3. Examples

### § 3.1 Variable Binding Results Examples

An example SELECT SPARQL Query in example.rq operating on query graph Turtle/N3 data in data.ttl providing ordered variable binding query results written in XML in output.srx.

This XML can be transformed into XHTML using the sample XML Query script result-to-html.xq giving output-xquery.html or with XSLT sheet result-to-html.xsl giving output-xslt.html

### § 3.2 Variable Binding Results Examples with Triple Terms

An example SELECT SPARQL Query in example-triple-terms.rq operating on query graph Turtle/N3 data in data.ttl providing ordered variable binding query results written in XML in output-triple-terms.srx. These results contain triple terms.

### § 3.3 Boolean Results Examples

An example ASK SPARQL Query in example2.rq operating on query graph Turtle/N3 data in data.ttl provides a boolean query result written in XML in output2.srx.

This XML can be transformed into XHTML using the sample XML Query script result-to-html.xq giving output-xquery2.html or with XSLT sheet result-to-html.xsl giving output-xslt2.html

## § 4. XML Schemas

There are normative XML schemas provided in the following formats:

1. RELAX NG [RELAXNG] Compact [RELAXNG-COMPACT] in result.rnc
2. RELAX NG XML in result.rng
3. W3C XML Schema [XMLSCHEMA-1] in result.xsd
   Note: this schema is machine-generated from the RELAX NG XML schema.

If W3C XML Schema is used, an `xsi:schemaLocation` attribute can be used pointing to the schema as follows:

```
<?xml version="1.0"?>
<sparql xmlns="http://www.w3.org/2005/sparql-results#"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="http://www.w3.org/2005/sparql-results

  ...

</sparql>
```

## § A. Changes between SPARQL Query Results XML Format (Second Edition) and SPARQL 1.2 Query Results XML Format

*This section is non-normative*.

- The document title changed from "SPARQL Query Results XML Format (Second Edition)" to "SPARQL 1.2 Query Results XML Format"
- Add RNC, RNG, and XSD files
- Allow triple terms to be expressed in 2.3.1 Variable Binding Results
- Use Media Type language instead of MIME Type in B. Internet Media Type, File Extension and Macintosh File Type
- Support directional language-tagged strings in 2.3.1 Variable Binding Results

## § B. Internet Media Type, File Extension and Macintosh File Type

The Internet Media Type (formerly known as MIME Type) for the SPARQL Query Results XML Format is "application/sparql-results+xml".

It is recommended that result files have the extension ".srx" (all lowercase) on all platforms.

It is recommended that result files stored on Macintosh HFS file systems be given a file type of "TEXT".

## § Internet Media Type Registration Form

**Type name:**
application

**Subtype name:**
sparql-results+xml

**Required parameters:**
None

**Optional parameters:**
"charset": This parameter has identical semantics to the charset parameter of the "application/xml" media type as specified in [RFC3023], section 3.2.

**Encoding considerations:**
Identical to those of "application/xml" as specified in [RFC3023], section 3.2.

**Security considerations:**

SPARQL query results uses URIs. See Section 7 of [RFC3986].

SPARQL query results uses IRIs. See Section 8 of [RFC3987].

As this media type uses the "+xml" convention, it shares the same security considerations as described in [RFC3023], section 10.

**Interoperability considerations:**

There are no known interoperability issues.

**Published specification:**

This specification.

**Applications which use this media type:**

No known applications currently use this media type.

**Additional information:**
**Magic number(s):**

As specified for "application/xml" in [RFC3023], section 3.2.

**File extension(s):**

".srx"

**Fragment identifiers:**

Identical to that of "application/xml" as described in [RFC3023], section 5.

**Base URI:**

As specified in [RFC3023], section 6.

**Macintosh file type code(s):**

"TEXT"

**Person & email address to contact for further information:**

Dave Beckett, Eric Prud'hommeaux <public-rdf-dawg-comments@w3.org>

**Intended usage:**

COMMON

**Restrictions on usage:**

None

**Author/Change controller:**

The SPARQL specification is a work product of the World Wide Web Consortium's RDF Data Access Working Group. The W3C has change control over these specifications.

## § C. Privacy Considerations

TODO

## § D. Security Considerations

TODO

## § E. Internationalization Considerations

TODO

## § F. Index

### § F.1 Terms defined by this specification

### § F.2 Terms defined by reference

[RDF12-CONCEPTS] defines the following:

RDF Terms

[SPARQL12-QUERY] defines the following:

Query Solution

Query Variable

Solution Sequence

SPARQL Query section 10

SPARQL Query section 10.2

SPARQL Query section 10.5

[XML] defines the following:

XML

# G. References

## G.1 Informative references


**[RDF12-CONCEPTS]**

*RDF 1.2 Concepts and Abstract Syntax*. Olaf Hartig; Pierre-Antoine Champin; Gregg Kellogg; Andy Seaborne. W3C. 19 December 2024. W3C Working Draft. URL: https://www.w3.org/TR/rdf12-concepts/

**[RELAXNG]**

*RELAX NG Specification*. James Clark; Murata Makoto. OASIS. 3 December 2001. URL: http://www.oasis-open.org/committees/relax-ng/spec-20011203.html

**[RELAXNG-COMPACT]**

*RELAX NG Compact Syntax*. James Clark. OASIS. 21 November 2002. URL: http://www.oasis-open.org/committees/relax-ng/compact-20021121.html

**[RFC3023]**

*XML Media Types*. M. Murata; S. St. Laurent; D. Kohn. IETF. January 2001. Proposed Standard. URL: https://www.rfc-editor.org/rfc/rfc3023

**[RFC3986]**

*Uniform Resource Identifier (URI): Generic Syntax*. T. Berners-Lee; R. Fielding; L. Masinter. IETF. January 2005. Internet Standard. URL: https://www.rfc-editor.org/rfc/rfc3986

**[RFC3987]**

*Internationalized Resource Identifiers (IRIs)*. M. Duerst; M. Suignard. IETF. January 2005. Proposed Standard. URL: https://www.rfc-editor.org/rfc/rfc3987

**[SPARQL12-CONCEPTS]**

*SPARQL 1.2 Concepts*. The W3C RDF-star Working Group. W3C. W3C Working Draft. URL: https://w3c.github.io/sparql-concepts/spec/

**[SPARQL12-ENTAILMENT]**

*SPARQL 1.2 Entailment Regimes*. Peter Patel-Schneider. W3C. 19 December 2024. W3C Working Draft. URL: https://www.w3.org/TR/sparql12-entailment/

**[SPARQL12-FEDERATED-QUERY]**

*SPARQL 1.2 Federated Query*. Ruben Taelman; Gregory Williams. W3C. 19 December 2024. W3C Working Draft. URL: https://www.w3.org/TR/sparql12-federated-query/

**[SPARQL12-GRAPH-STORE-PROTOCOL]**

*SPARQL 1.2 Graph Store Protocol*. Andy Seaborne; Thomas Pellissier Tanon. W3C. 19 December 2024. W3C Working Draft. URL: https://www.w3.org/TR/sparql12-graph-store-protocol/

**[SPARQL12-NEW]**

*What's New in SPARQL 1.2*. The W3C RDF-star Working Group. W3C. W3C Working Draft. URL: https://w3c.github.io/sparql-new/spec/

**[SPARQL12-PROTOCOL]**

*SPARQL 1.2 Protocol*. Andy Seaborne; Ruben Taelman; Gregory Williams; Thomas Pellissier Tanon. W3C. 19 December 2024. W3C Working Draft. URL: https://www.w3.org/TR/sparql12-protocol/

**[SPARQL12-QUERY]**

*SPARQL 1.2 Query Language*. Olaf Hartig; Andy Seaborne; Ruben Taelman; Gregory Williams; Thomas Pellissier Tanon. W3C. 19 December 2024. W3C Working Draft. URL: https://www.w3.org/TR/sparql12-query/

**[SPARQL12-RESULTS-CSV-TSV]**

*SPARQL 1.2 Query Results CSV and TSV Formats*. Ruben Taelman; Gregory Williams; Thomas Pellissier Tanon. W3C. 19 December 2024. W3C Working Draft. URL: https://www.w3.org/TR/sparql12-results-csv-tsv/

**[SPARQL12-RESULTS-JSON]**

*SPARQL 1.2 Query Results JSON Format*. Andy Seaborne; Ruben Taelman; Gregory Williams; Thomas Pellissier Tanon. W3C. 19 December 2024. W3C Working Draft. URL: https://www.w3.org/TR/sparql12-results-json/

**[SPARQL12-SERVICE-DESCRIPTION]**

*SPARQL 1.2 Service Description*. Ruben Taelman; Gregory Williams. W3C. 19 December 2024. W3C Working Draft. URL: https://www.w3.org/TR/sparql12-service-description/

**[SPARQL12-UPDATE]**

*SPARQL 1.2 Update*. Ruben Taelman; Andy Seaborne; Thomas Pellissier Tanon. W3C. 19 December 2024. W3C Working Draft. URL: https://www.w3.org/TR/sparql12-update/

**[XML]**

*Extensible Markup Language (XML) 1.0 (Fifth Edition)*. Tim Bray; Jean Paoli; Michael Sperberg-McQueen; Eve Maler; François Yergeau et al. W3C. 26 November 2008. W3C Recommendation. URL: https://www.w3.org/TR/xml/

**[XMLSCHEMA-1]**

*XML Schema Part 1: Structures Second Edition*. Henry Thompson; David Beech; Murray Maloney; Noah Mendelsohn et al. W3C. 28 October 2004. W3C Recommendation. URL: https://www.w3.org/TR/xmlschema-1/

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